CS Lab

Laboratories

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CHAPTER 1

Algorithms

First of all, what is an algorithm?

The word itself has an origin story that traces back to Persian scientist and polymath Al-Khwarizmi. Al-Khwarizmi's work on arithmetic introduced systematic calculation methods and the Arabic numeral system to medieval Europe. He also presented clear, step-by-step procedures for solving common arithmetic and algebraic problems. The world "algorithm" is derived from the latinised form of his name.

An algorithm itself is a sequence of well-defined instructions used to perform a particular task. Furthermore, an algorithm may receive an input, process it, and deliver an output. A complete algorithm must terminate after a finite number of steps.

Having covered the words origin and meaning, the rest of this chapter aims to introduce some of the most important algorithms and techniques in computer science.

1.1. Techniques

TODO

1.1.1. Iteration

TODO

1.1.2. Recursion

TODO

1.1.3. Divide and conquer

TODO

1.1.4. Greedy algorithms

TODO

1.1.5. Backtracking

TODO

1.1.6. Heuristic

1.2. Sorting

One of the most frequently recurring tasks in day-to-day lives is the sorting of elements. Regardless of what is being sorted, there is always a systematic approach behind the sorting process.

For example, when sorting a list of numbers ascendingly, one could repeatedly find the smallest number in the unsorted part of the list and swap it with the first number in that part. This approach is known as selection sort or more specifically, min sort.

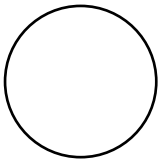
1.2.1. The problem

Imagine you are handed a large pile of documents that have to be arranged according to their page numbers. While sorting a small number of papers by hand sounds feasible, the process becomes increasingly time-consuming and error-prone as the number of pages increases. Therefore, a systematic sorting method is needed. Two commonly used methods for solving this type of problem are Merge Sort (Section 1.2.2) and Quick Sort (Section 1.2.3).

1.2.2. Merge Sort

Merge Sort utilises the “divide-and-conquer” principle to solve the problem presented above. First, the given list has to be recursively split into two halves until each resulting sublist contains exactly one element. The sublists are then merged back in sorted order.

Let’s imagine this is what our list looks like:



1.2.3. Quick Sort

TODO